

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

PRACTICAL LAB 1

Linux Filesystem, Paths and File Manipulation

A practical evidence-based laboratory report

Hands-on practice with VFS concepts, inodes, directory navigation, file creation, copying, moving, deletion and text search

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Task 1: Orient Yourself in the Linux Filesystem

The terminal evidence below demonstrates the initial working directory, ordinary and hidden entries, movement to the filesystem root, and return to the home directory.

```
(tempadmin@kali) - [~]  
$ pwd  
/home/tempadmin  
  
(tempadmin@kali) - [~]  
$ ls  
Desktop  Downloads  Pictures  Public  Videos  
Documents Music      Projects  Templates  
  
(tempadmin@kali) - [~]  
$ ls -a  
.  
..  
.bash_history  
.bash_logout  
.bashrc  
.bashrc.original  
.cache  
.config  
Desktop  
Documents  
Downloads  
.java  
.local  
Music  
Pictures  
.profile  
Projects  
Public  
.sudo_as_admin_successful  
Templates  
Videos  
.zprofile  
.zshrc  
  
(tempadmin@kali) - [~]  
$ cd /  
  
(tempadmin@kali) - [/  
$ ls  
bin  etc  initrd.img  lib32  media  proc  sbin  this  var  
boot got  initrd.img.old  lib64  mnt   root  srv   tmp   vmlinuz  
dev  home lib          lost+found  opt   run   sys   usr   vmlinuz.old  
  
(tempadmin@kali) - [/  
$ cd ~  
  
(tempadmin@kali) - [~]  
$ pwd  
/home/tempadmin  
  
(tempadmin@kali) - [~]  
$
```

Figure 1.0 — Working-directory and listing checks, including the return to the home directory.

Written response: The root directory, written as /, is the top of the Linux filesystem hierarchy and contains system-wide directories such as /etc, /usr, and /home. The home directory is the personal workspace assigned to a user; in this evidence it is shown as /home/tempadmin, where the user can normally create and manage personal files.

Task 2: Inspect Inodes and File Metadata

An inode stores metadata about a filesystem object, including ownership, permissions, timestamps, size, and the filesystem reference used to locate its data. The ls -li output was captured before and after renaming alpha.txt.

```
(tempadmin@kali) - [~]
$ mkdir -p ~/icdfa_lab1

(tempadmin@kali) - [~]
$ cd ~/icdfa_lab1/

(tempadmin@kali) - [~/icdfa_lab1]
$ touch alpha.txt beta.txt

(tempadmin@kali) - [~/icdfa_lab1]
$ ls -li
total 0
1230236 -rw-rw-r-- 1 tempadmin tempadmin 0 Sep  7 15:02 alpha.txt
1230237 -rw-rw-r-- 1 tempadmin tempadmin 0 Sep  7 15:02 beta.txt

(tempadmin@kali) - [~/icdfa_lab1]
$ mv alpha.txt alpha_renamed.txt

(tempadmin@kali) - [~/icdfa_lab1]
$ ls -li
total 0
1230236 -rw-rw-r-- 1 tempadmin tempadmin 0 Sep  7 15:02 alpha_renamed.txt
1230237 -rw-rw-r-- 1 tempadmin tempadmin 0 Sep  7 15:02 beta.txt

(tempadmin@kali) - [~/icdfa_lab1]
$
```

Figure 2.0 — Inode numbers before and after renaming `alpha.txt` to `alpha_renamed.txt`.

Observation: The inode number did not change after the rename. The evidence shows inode 12386 for `alpha.txt` before the operation and the same inode 12386 for `alpha_renamed.txt` afterwards; the filename changed, but the underlying filesystem object remained the same.

Task 3: Practice Absolute and Relative Paths

The same file was accessed from two different locations. This demonstrates that a path may identify an object from the root of the hierarchy or relative to the current working directory.

A terminal window showing a series of commands and their outputs. The prompt is `(tempadmin@kali) - [~/icdfa_lab1]`. The commands and outputs are:
1. `$ mkdir ~/icdfa_lab1/folder-1 ~/icdfa_lab1/folder-2`
2. `$ echo "ICDFA Linux Path Practice" > ~/icdfa_lab1/folder-1/file.txt`
3. `$ cat ~/icdfa_lab1/folder-1/file.txt`
Output: `ICDFA Linux Path Practice`
4. `$ cd ~/icdfa_lab1/folder-2`
5. `$ pwd`
Output: `/home/tempadmin/icdfa_lab1/folder-2`
6. `$ cat ../folder-1/file.txt`
Output: `ICDFA Linux Path Practice`
7. `$ cd ..`
8. `$ pwd`
Output: `/home/tempadmin/icdfa_lab1`
9. `$`
Annotations: A red circle with the number '1' is placed over the `cd` command in step 4. A red circle with the number '2' is placed over the `pwd` command in step 8. Red arrows point from these circles to the right.

Figure 3.0 — Successful access to `file.txt` using both an absolute path and a relative path.

Written response: An absolute path begins at the root directory and identifies a location independently of the current directory; for example, ~/icdfa_lab1/folder-1/file.txt expands to the user's home-based full path. A relative path is interpreted from the current directory; while inside folder-2, ../folder-1/file.txt means "go up one level, then enter folder-1 and read file.txt."

Task 4: Explore the PATH Variable

PATH is a colon-separated list of directories searched by the shell when a command is entered without an explicit path. The evidence identifies the ls executable and shows the temporary mybin directory added to the current session.



```
(tempadmin@kali) - [~/icdfa_lab1]
$ echo "$PATH"
/home/tempadmin/.local/bin:/usr/local/sbin:/usr/sbin:/sbin:/usr/local/bin:/usr/bin:/bin:/usr
tools

(1) (tempadmin@kali) - [~/icdfa_lab1]
$ which ls
/usr/bin/ls

(tempadmin@kali) - [~/icdfa_lab1]
$ ls -l $(which ls)
-rwxr-xr-x 1 root root 166792 Feb 26 2026 /usr/bin/ls

(tempadmin@kali) - [~/icdfa_lab1]
$ mkdir ~/icdfa_lab1/mybin

(tempadmin@kali) - [~/icdfa_lab1]
$ export PATH="$HOME/icdfa_lab1/mybin:$PATH"

(tempadmin@kali) - [~/icdfa_lab1]
$ echo "$PATH"
/home/tempadmin/icdfa_lab1/mybin:/home/tempadmin/.local/bin:/usr/local/sbin:/usr/sbin:/sbin:/usr/games:/home/tempadmin/.dotnet/
(2) (tempadmin@kali) - [~/icdfa_lab1]
$
```

Figure 4.0 — which ls result and the modified PATH value containing ~/icdfa_lab1/mybin.

Why ls works from different directories: The shell searches each directory listed in PATH until it finds an executable named ls. Because /usr/bin is in PATH, the command can be entered as ls from any working directory instead of typing /usr/bin/ls.

If a command cannot be found: If no executable with that name exists in any PATH directory, the shell reports an error such as "command not found" and does not execute the command. The user must correct the command name, provide the executable's path, or add the correct directory to PATH.

Task 5: Create, View, Copy, Move and Delete Files

Each operation was performed inside the lab workspace and checked through subsequent terminal output. The final listing confirms that the original file remained in practice and the backup copy remained in backup after the moved file and recursive copy were removed.

```

(tempadmin@kali)-[~/icdfa_lab1]
$ mkdir -p ~/icdfa_lab1/practice

(tempadmin@kali)-[~/icdfa_lab1]
$ echo "hello world" > ~/icdfa_lab1/practice/my_first_file.txt

(tempadmin@kali)-[~/icdfa_lab1]
$ cat ~/icdfa_lab1/practice/my_first_file.txt
hello world
1

(tempadmin@kali)-[~/icdfa_lab1]
$ cp ~/icdfa_lab1/practice/my_first_file.txt ~/icdfa_lab1/practice/copy.txt

(tempadmin@kali)-[~/icdfa_lab1]
$ mkdir -p ~/icdfa_lab1/backup

(tempadmin@kali)-[~/icdfa_lab1]
$ cp ~/icdfa_lab1/practice/my_first_file.txt ~/icdfa_lab1/backup

(tempadmin@kali)-[~/icdfa_lab1]
$ cp -r ~/icdfa_lab1/practice ~/icdfa_lab1/practice_copy
2

(tempadmin@kali)-[~/icdfa_lab1]
$ mv ~/icdfa_lab1/practice/copy.txt ~/icdfa_lab1/practice/renamed_copy.txt

(tempadmin@kali)-[~/icdfa_lab1]
$ mv ~/icdfa_lab1/practice/renamed_copy.txt ~/icdfa_lab1/backup

(tempadmin@kali)-[~/icdfa_lab1]
$ rm ~/icdfa_lab1/backup/beta.txt

(tempadmin@kali)-[~/icdfa_lab1]
$ rm ~/icdfa_lab1/backup/renamed_copy.txt

(tempadmin@kali)-[~/icdfa_lab1]
$ rm -r ~/icdfa_lab1/practice_copy/

(tempadmin@kali)-[~/icdfa_lab1]
$ ls -l ~/icdfa_lab1/practice/
total 4
-rw-rw-r-- 1 tempadmin tempadmin 12 Sep  8 03:22 my_first_file.txt
3

(tempadmin@kali)-[~/icdfa_lab1]
$ ls -l ~/icdfa_lab1/backup
total 4
-rw-rw-r-- 1 tempadmin tempadmin 12 Sep  8 03:26 my first file.txt
4

(tempadmin@kali)-[~/icdfa_lab1]
$

```

Figure 5.0 — File creation and viewing, copying, recursive copying, rename/move operations, deletion, and final directory verification.

Verification summary: The original my_first_file.txt was created and displayed, copied to copy.txt, backed up, and recursively copied as practice_copy. copy.txt was renamed to renamed_copy.txt and moved to backup; it was then removed, as was practice_copy. The final listings show my_first_file.txt in practice and a backup copy in backup.

Task 6: Search Text with grep

grep searches text streams or files for lines matching a pattern. The -n option adds line numbers, which makes the result easier to verify and reference.

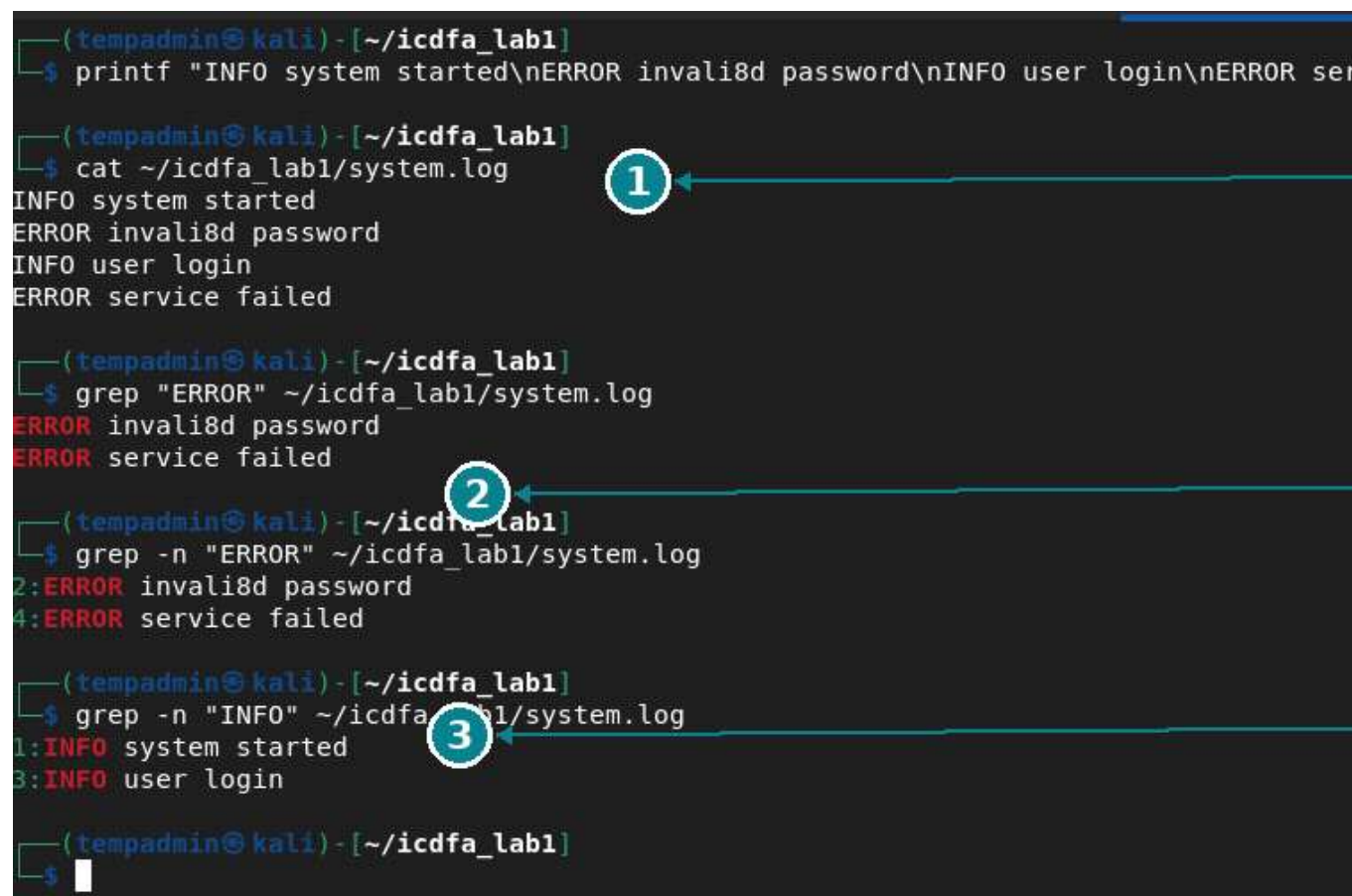
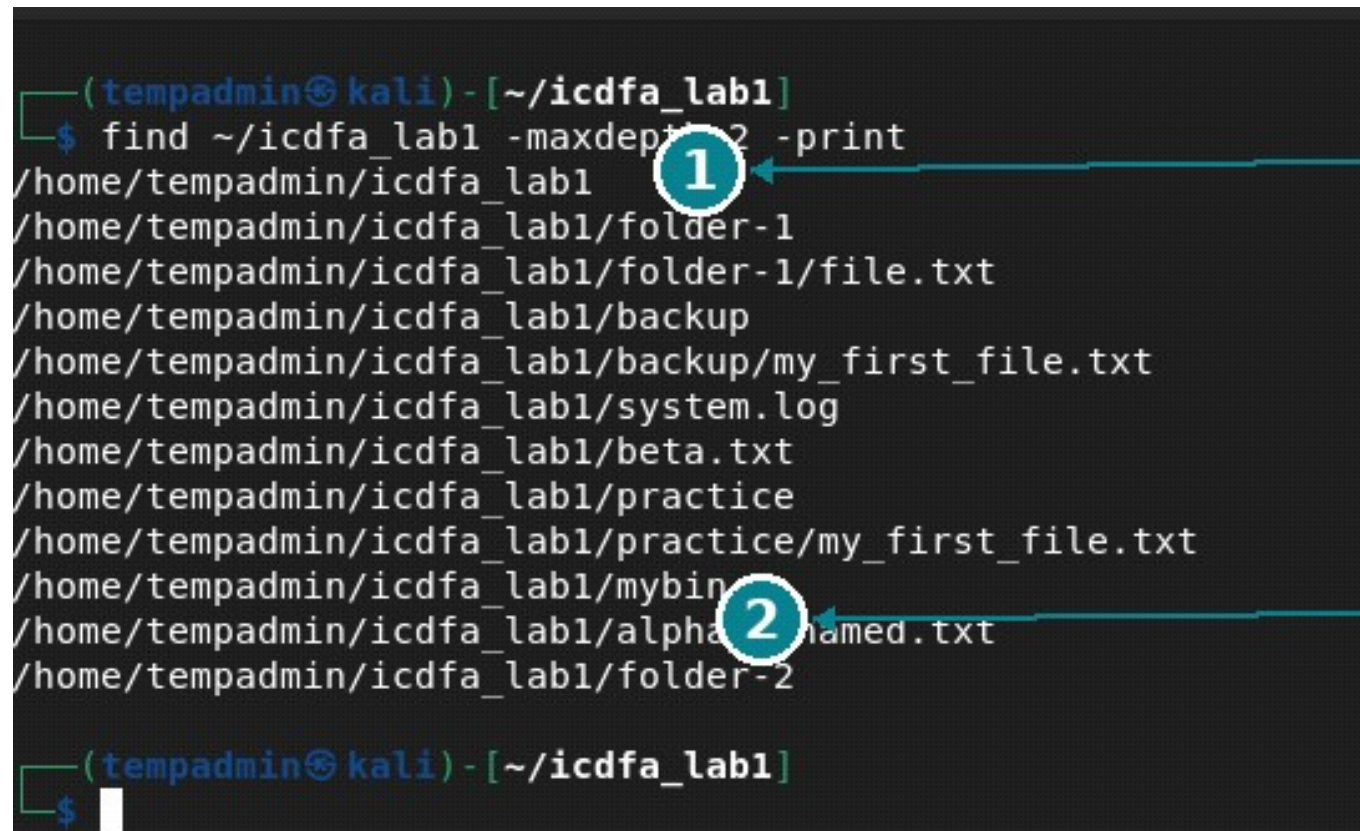
A terminal window showing a series of commands and their outputs. The terminal is in a Kali Linux environment with the user 'tempadmin' in the directory '~/icdfa_lab1'. The commands and outputs are as follows:
1. Command: `printf "INFO system started\nERROR invalid password\nINFO user login\nERROR service failed"`
2. Command: `cat ~/icdfa_lab1/system.log`
 Output: `INFO system started`
 `ERROR invalid password`
 `INFO user login`
 `ERROR service failed`
3. Command: `grep "ERROR" ~/icdfa_lab1/system.log`
 Output: `ERROR invalid password`
 `ERROR service failed`
4. Command: `grep -n "ERROR" ~/icdfa_lab1/system.log`
 Output: `2:ERROR invalid password`
 `4:ERROR service failed`
5. Command: `grep -n "INFO" ~/icdfa_lab1/system.log`
 Output: `1:INFO system started`
 `3:INFO user login`
The output of the third command is highlighted in red. Red arrows with circular numbers 1, 2, and 3 point to the first, second, and third command lines respectively.

Figure 6.0 — Complete system.log contents and grep results for ERROR and INFO with line numbers.

Practical cybersecurity/system-administration use: An administrator can use grep to review logs for indicators such as failed logins, permission denials, malware-related strings, or service errors. For example, `grep -n "ERROR" system.log` quickly isolates error events and preserves their line numbers for investigation, ticketing, or correlation with other timestamps and logs.

Task 7: Cleanup and Reflection

The find command was used to display the complete lab directory tree to a maximum depth of two before final cleanup. This is a useful final check because it confirms which files and directories remain and prevents accidental deletion of the wrong location.



```
(tempadmin@kali) - [~/icdfa_lab1]
$ find ~/icdfa_lab1 -maxdepth 2 -print
/home/tempadmin/icdfa_lab1
/home/tempadmin/icdfa_lab1/folder-1
/home/tempadmin/icdfa_lab1/folder-1/file.txt
/home/tempadmin/icdfa_lab1/backup
/home/tempadmin/icdfa_lab1/backup/my_first_file.txt
/home/tempadmin/icdfa_lab1/system.log
/home/tempadmin/icdfa_lab1/beta.txt
/home/tempadmin/icdfa_lab1/practice
/home/tempadmin/icdfa_lab1/practice/my_first_file.txt
/home/tempadmin/icdfa_lab1/mybin
/home/tempadmin/icdfa_lab1/alpha_named.txt
/home/tempadmin/icdfa_lab1/folder-2

(tempadmin@kali) - [~/icdfa_lab1]
$
```

Figure 7.0 — Final lab-directory tree displayed with find before cleanup.

What is an inode? An inode is a filesystem data structure that stores metadata about a file or directory and identifies its associated data blocks. It does not normally store the filename; directory entries map filenames to inode numbers.

Why is pwd important? pwd prints the current working directory. Checking it before file manipulation reduces the risk of creating, moving, or deleting items in the wrong location, especially when relative paths are used.

Difference between cp and mv: cp creates a second copy while leaving the original in place. mv changes the location or name of the same filesystem object; when used within one filesystem, it generally changes the directory entry rather than duplicating file data.

Why rm must be used carefully: rm normally removes files without moving them to a recycle bin, so recovery may be difficult or impossible. A mistaken path, wildcard, or recursive

option can remove important data; the target should therefore be checked carefully before execution.